
STEEL FAB ENTERPRISES (SFE) MILESTONE TRACKING SYSTEM - WORKFLOW DOCUMENT

This document details the complete functional architecture, workflows, modules, and button-by-button operations of the SFE Milestone Tracking and Estimation Portal.

Use this file as reference context to train or ground chatbot assistance workflows.

1. GLOBAL INTERFACE WORKFLOWS & NAVIGATION

* NAVIGATION SIDEBAR (Layout/Sidebar.jsx):

Located on the left side of the screen. Supports Desktop collapsing to compact mode and Mobile sliding responsive navigation drawer. Contains sections:

- OVERVIEW: System dashboards.
- BID MANAGEMENT: RFQ logging, calendars, schedules, and estimation tabs.
- MANAGEMENT: Directories for Employees, Customers, and Detailers.
- BUDGET ESTIMATOR: Quick structural budget estimators.
- PROJECT MANAGEMENT: Detailed projects and milestones master lists.
- STRUCTURAL SCHEDULE: Detailing plan creation and Gantt timeline tracking.
- PRODUCTION MANAGEMENT: Priority scheduling and routing process master.
- CAPACITY MAPPING: Resource configuration, machine master, and workforce shifts.
- SYSTEM: Settings and global announcements.

* NAVIGATION HEADER (Layout/Header.jsx):

Located at the top of the main content pane.

- "Menu" icon (lucide Menu): Toggles sidebar drawer on mobile.
- "Back" button (lucide ArrowLeft): Returns user to previous URL.
- Centered Title: Dynamically displays active module name.
- Clock/Date indicator: Displays real-time local date and time.
- "Notifications" bell (lucide Bell): Toggles a dropdown pane displaying alerts.
- Profile Avatar: Displays current user initials/image. Toggles a menu to logout.
- "Sign Out" button: Clears session credentials and returns user to login page.

* FLOATING ASSISTANT WIDGET (Chatbot/ChatbotWidget.jsx):

- Floating Orange Icon (bottom-right corner): Toggles the Chatbot pane.
- Settings (Gear) Icon in header: Opens Context Manager. Allows admins to upload workflow documents. Displays offline Ollama connection and model statuses.
- "New" Button (+ symbol in sidebar): Resets the context and starts a new session.
- Citation tags: Lists page numbers from which RAG information was sourced.

2. OVERVIEW MODULE

* DASHBOARD:

Aggregates overall statistics for bids, projects, timelines, and announcements.

- Cards: High-level numbers of active bids, late projects, and pending RFQs.
- Announcement list: Displays system-wide alerts broadcast by administrators.

3. BID MANAGEMENT MODULES

* RFQ MASTER (rfq/):

Log and process Request for Quotation documents.

- "Add RFQ" button: Opens a form modal to input client name, date received, due date, estimators assigned, and scope of work details.
- Estimators Assignee grid: Assign estimators, set estimator levels.
- Dollar Dashboard: Evaluates historical bid amounts, average values, win rates.
- Job Analytics: Evaluates win/loss metrics grouped by sector, scale, and client.
- Print Setup: Exports data sheets to printable formats.

* INTERNAL BID SCHEDULE:

Tracks critical internal milestones before bid submission.

- Calendar View: Interactive calendar marking upcoming estimate review meetings.
- List View: Filterable lists of review dates.
- "Add Entry" button: Schedules estimator review sessions, assignees, dates.

* HOLIDAY CALENDAR:

Central log of non-working days used to adjust estimation/production schedules.

- "Add Holiday" button: Opens a modal with Name, Start Date, and End Date fields.
- Table: Lists holidays with "Edit" (pencil) and "Delete" (trashcan) controls.

* ESTIMATION & ERECTION (estimation-erection/):

The primary module for structural engineering weight and cost estimations.

Contains 6 main tabs inside EstimationErectionLayout:

1. CORE TAB (CoreTab.jsx):

- Calculates core structural formulas and coefficients.
- Controls: Fields for base parameters (e.g. building dimensions, spans).
- Calculation buttons: Recalculate weights, run formulas.

2. BREAKDOWN TAB (BreakdownTab.jsx):

- Breaks down structural steel elements into component categories (columns, beams, bracing, trusses).
- Controls: Editable tables with weight rows, save coefficients.

3. ERECTION TAKEOFF TAB (ErectionTakeoffTab.jsx):

- Computes the costs associated with hoisting and erecting the structure.
- Input fields: Crane tonnage, rental days, rigging crew size, travel rates.
- "Recalculate Erection Costs" button: Computes final erection prices.

4. FIELD MOMENT CONNECTION TAB (FieldMomentConnTab.jsx):

- Calculates joint welding, bolting, and splice connection requirements.
- Fields: Bolt diameters, connection counts, field weld lengths.
- Calculation controls: Updates detailing hours and material weight add-ons.

5. MISC METALS TAB (MiscMetalsTab.jsx):

- Inputs for secondary steel members (staircases, handrails, ladders, gratings).
- Controls: Unit inputs (tons/meters), unit costs.
- Buttons: Saves miscellaneous metals components breakdown.

6. ESTIMATE DATA TAB (EstimateDataTab.jsx):

- Reference table of master material rates, detailing rates, and shop labor rates.
- Controls: Edit rates inline, click "Save Rates" to commit updates.

4. DIRECTORY MANAGEMENT MODULES

* EMPLOYEE MASTER:

Database of shop technicians, estimators, and managers.

- "Add Employee" button: Opens a modal with fields: Name, Role, Email, Phone.
- Filters: Quick filtering by Role (Estimator, Shop Staff, Administrator).
- Actions: Edit profile details, delete employee records.

* CUSTOMER MASTER:

Database of structural contracting clients.

- "Add Customer" button: Modal fields: Company Name, Contact Person, Email, Rating.
- Search bar: Real-time search by Company Name.
- Actions: Edit Customer profile, Delete records.

*** DETAILER MASTER:**

Database of external engineering detailers who translate concepts to shop drawings.

- "Add Detailer" button: Modal fields: Detailer Name, Specialization, Capacity.
- Actions: Edit capacity details, Delete records.

5. BUDGET ESTIMATOR MODULE

*** DESIGN INPUTS:**

- Form: Enter width, length, height, roof slope, and frame spacing specifications.
- "Calculate Budget" button: Sends data to estimation rules engine.

*** ESTIMATION RESULT:**

- Tonnage summary: Shows computed structural weights in tons.
- Cost summary: Estimates fabrication labor, material, and logistics cost ranges.
- "Export PDF" button: Generates a proposal budget sheet.

6. PROJECT & MILESTONE MANAGEMENT MODULES

*** PROJECT MASTER:**

Main repository of ongoing construction projects.

- "Add Project" button: Modal fields: Project Name, Customer, Start/End Dates.
- Project Cards: Displays active projects, visual progress meters.

- "Manage Milestones" link: Navigates directly to the project's milestones list.

* MILESTONE MANAGEMENT:

Checks deliverables and deadlines.

- "Add Milestone" button: Modal fields: Title, Project, Assignee, Due Date.
- Filter Tabs (All, Completed, In Progress, Overdue, Pending): Quick toggles.
- "Edit" icon (pencil): Updates milestone status (e.g. from Pending to In Progress).
- "Delete" icon (trashcan): Deletes milestone.

7. STRUCTURAL SCHEDULE MODULES

* PLAN CREATION:

Sets sequence and releases detailing phases to production.

- "Create Phase" button: Defines structural sequence names and boundaries.
- Input tables: Map tonnages to specific detailing zones.

* PLAN TRACKING:

- Gantt Chart (GanttChart.jsx): Visualization of phase timelines.
- Timeline slider: Zoom in/out of weekly/monthly timeline views.
- Progress checks: Marks detailing, fabrication, and shipping steps.

8. PRODUCTION MANAGEMENT MODULES

* PRODUCTION PRIORITY SCHEDULE:

Tracks raw materials through shop processes (Plate, Angle, Structural).

- Prioritize Table: Set priority ranks for components.
- "Update Schedule" button: Updates production release sequencing.
- Status Indicators: Displays material stages (Not Started, Cutting, Assembled, Welded, Painted, Loaded).

* PROCESS MASTER:

Configures fabrication shop stations.

- "Add Process Station" button: Defines new routing steps (e.g., CNC Plasma, Shotblast, Priming).
- Station lists: Shows routing sequence patterns.

9. CAPACITY MAPPING MODULES

* CAPACITY CONFIGURATION:

- Form: Inputs monthly output targets (in tons).
- "Save Config" button: Updates baseline efficiency variables.

* MACHINE MASTER:

- "Add Machine" button: Modal fields: Machine Name, Station, Efficiency Rating.
- Machine table: Monitors downtime and capacity constraints.

* WORKFORCE MASTER:

- "Add Shift Allocation" button: Modal fields: Employee, Shift (Morning, Night, Swing), Allocated station.
- Allocation grid: Prevents resource double-booking.

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